Student Management System in C++

This C++ code implements a Student Management System where you can add, display, and delete student records. Below is a detailed explanation, breaking down its structure and logic for a mini-project exam:

1. Includes and Libraries

- **iostream**: For input/output operations.
- **conio.h**: For functions like **getch()** to handle input and output without the need to press Enter.
- **cstdlib**: For general utilities, though not fully used here.
- **fstream**: To handle file input/output for storing and reading student records.
- **windows.h**: For manipulating console output, like changing text colors using **SetConsoleTextAttribute**.

2. Set Color Function

- This function sets the text and background color in the console using Windows API. It uses color codes where foreground and background are integers representing color values.

3. Student Class

The student class holds attributes and methods for managing student information.

Private Attributes:

- **s_name, m_name, f_name, c_name, cr_name**: Personal details (student name, parent names, college, course).
- **roll_no, rgln, dd, mm, yy**: Roll number, regulation, and birth date.
- **phy, chm, mat, P_S, ECM**: Marks in subjects.
- **per**: Percentage.
- **sum**: Total marks in all subjects.
- **grade, result**: Grade and result (PASS/FAIL).

Public Methods:

- **menu()**: Displays the main menu for the user to choose actions.
- **insert()**: For entering student records and saving them to a file.
- **display()**: To display the student records stored in the file.
- **deleted()**: To delete a student record from the file based on roll number.

4. Menu Function

- Displays a menu with options to add, display, or delete student records.
- Uses a loop to allow adding multiple records until the user chooses otherwise.

5. Insert Function

- Collects student details (name, marks, etc.) and stores them in a file **studentresult.txt**.
- Computes the total and percentage, assigns a grade based on percentage, and stores the result as either "PASS" or "FAIL".

6. Display Function

- Reads and displays all the student records from the file **studentresult.txt**.

7. Delete Function

- Reads the file and copies all records except the one with the matching roll number to a new file. The old file is deleted, and the new file is renamed.

8. Main Function

- The main function initializes a student object and calls the **menu()** function to start the program.

Key Points:

- **File handling**: The program uses **fstream** to read/write student data.
- **Color manipulation**: **SetConsoleTextAttribute** is used to change text color for better visual experience.
- **Menu navigation**: The program provides an interactive menu for user input.
- **Student details**: The system allows storing and displaying student details like name, marks, and results.
- **Error handling**: The program handles cases like no data found or incorrect roll number during deletion.

This code covers basic file operations and interactive menu-driven logic, making it suitable for a mini-project.